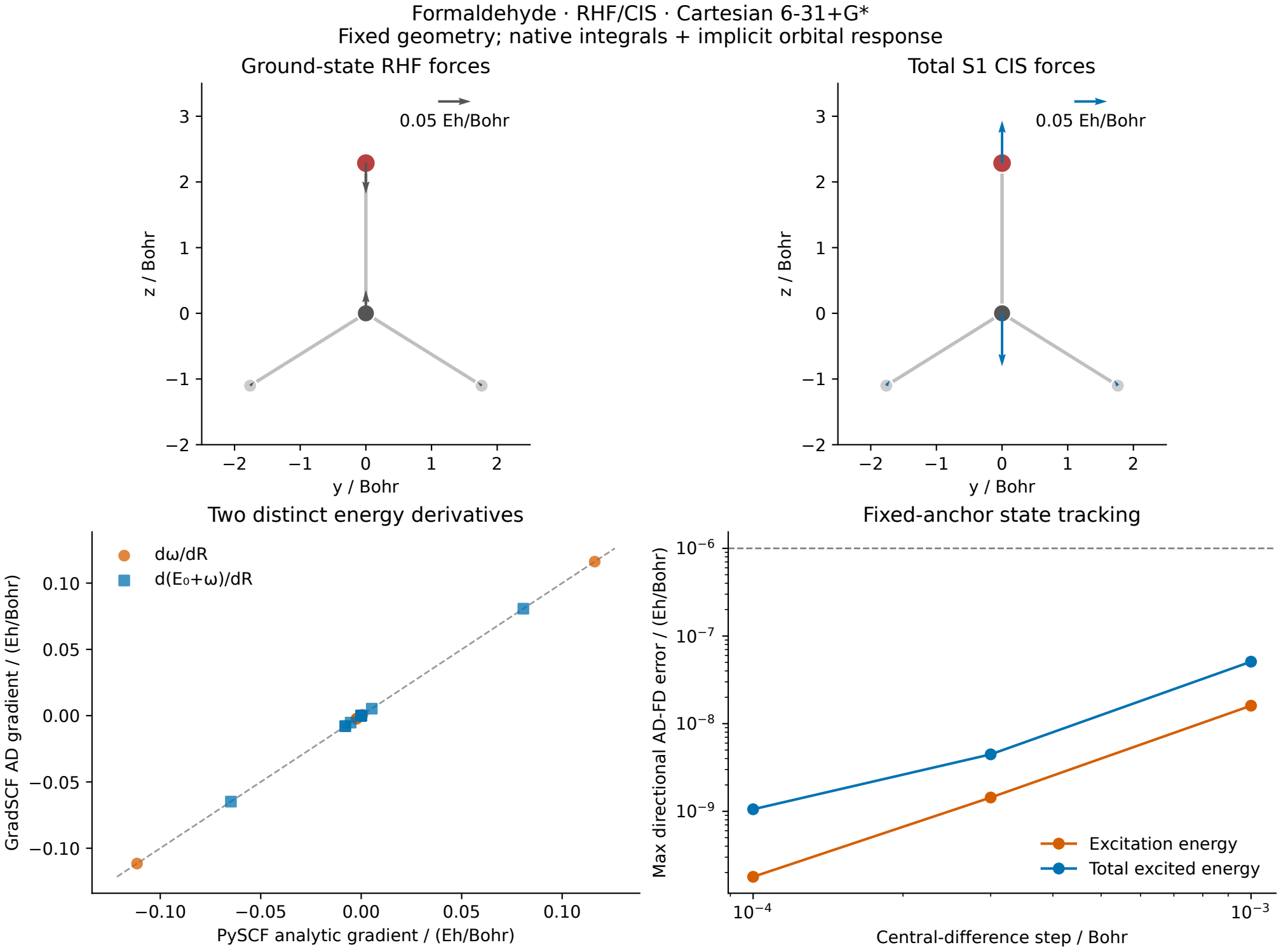
Formaldehyde · RHF/CIS · Cartesian 6-31+G*
Fixed geometry; native integrals + implicit orbital response
Ground-state RHF forces
0.05 Eh/Bohr
z / Bohr
y / Bohr
Total S1 CIS forces
0.05 Eh/Bohr
z / Bohr
y / Bohr
Two distinct energy derivatives
dω/dR
d(E₀+ω)/dR
GradSCF AD gradient / (Eh/Bohr)
PySCF analytic gradient / (Eh/Bohr)
Fixed-anchor state tracking
Max directional AD-FD error / (Eh/Bohr)
Excitation energy
Total excited energy
Central-difference step / Bohr